Contractive Centered L^p Decay Forces Poincaré Without Symmetry

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Status and source question

This is a `candidate_full_likely_valid` result for Patrick Cattiaux, Arnaud Guillin, and Cyril Roberto, *Poincaré inequality and the L^p convergence of semi-groups*, arXiv:1003.0784 [1]. Section 4 on PDF page 12 discusses an invariant, not necessarily symmetric, Markov diffusion semigroup. The authors assume that, for some $p > 2$, the centered decay estimate

$$N_p(P_t f) \le e^{-\lambda_p t} N_p(f), \qquad N_p(f) = \|f - \mu(f)\|_p, \tag{1}$$

holds with the crucial prefactor $K_p = 1$. They obtain an L^2 exponential estimate by duality and interpolation but state that, because they do not know that $K_2 = 1$, they cannot conclude the Poincaré inequality.

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4. Some final Remarks.

We did not succeed in proving the analogue of Lemma 1.2 for $p > 2$ (and actually we believe that such a statement is false). Hence both Theorems 1.4, 1.5 and 3.10 have their own interest.

Of course under stronger assumptions than the sole Poincaré inequality (logarithmic Sobolev inequality for instance), one can improve the bounds obtained in Theorem 1.1.

Extension to the non-symmetric case.

Notice that the only point where we used symmetry is the proof of Lemma 2.1, hence of Lemma 1.2. In particular if μ is invariant but not necessarily symmetric, (**H-Poinc**) implies exponential decay in all the $\mathbb{L}^p(\mu)$, $p \ge 2$, and our bounds are available, in particular we may choose $K_p = 1$.

But if (**H-p**) holds for some $p > 2$ and with $K_p = 1$ (which is crucial) then (2.4) is satisfied, which in return implies the same decay for the dual semi-group P_t^*. Hence the duality argument shows that (**H-q**) is satisfied for both P_t and P_t^*, where q is the conjugate exponent of p. Hence (**H-Poinc**) implies exponential decay in all the $\mathbb{L}^p(\mu)$, $1 < p < +\infty$.

Conversely, assume that (**H-p**) holds for some $p > 2$ and with $K_p = 1$ (which is still crucial). The previous argument shows that (**H-q**) is satisfied. The Riesz-Thorin interpolation theorem then shows that (**H-s**) is satisfied for all $q \le s \le p$, hence for $s = 2$. But since we do not know that $K_2 = 1$, we cannot conclude that the Poincaré inequality is satisfied. Also note that the induction argument we used in the proofs calls explicitly upon the Poincaré inequality, so that we cannot deduce that (**H-s**) holds for $s > p$.

Finally recall that in the non-symmetric situation, exponential decay in $\mathbb{L}^2$ can occur while the Poincaré inequality is not satisfied. Of course in this situation, $K_2 > 1$. This is the generic situation in many hypocoercive kinetic models like the kinetic Ornstein-Uhlenbeck process studied in [8] (also see [2] section 6).

The theorem below closes precisely this converse under the diffusion hypotheses used by the source.

Setting and proof intuition

Let $(P_t)_{t\geq 0}$ be a Markov semigroup with invariant probability measure μ, generator L, and carré du champ

$$\Gamma(f,g) = \frac{1}{2}\big(L(fg) - fLg - gLf\big).$$

Assume the source's diffusion chain rule, so $\Gamma(\phi(f),\phi(f)) = (\phi'(f))^2\Gamma(f,f)$ for smooth ϕ. Symmetry is not assumed.

Differentiating (1) at zero gives the weighted inequality that the source labels (2.4). The tempting substitution $h = \operatorname{sgn}(g)|g|^{2/p}$ makes its energy exactly quadratic, but the source correctly notes two problems: this map is not C^2 and h need not have mean zero.

Both are removed at once. Smooth the fractional power and first shift g by a scalar c_ε chosen so that the smoothed fractional power has mean zero. Its weighted energy multiplier is at most one, while its p-th power converges to $|g-c|^2$. The limiting c need not be the mean: variance is no larger than the squared distance from any constant.

Main result

Theorem 1 (nonsymmetric converse)**.** *Suppose the invariant Markov diffusion semigroup above satisfies* (1) *for some $p > 2$ and $\lambda_p > 0$, for every $f \in L^p(\mu)$ and $t \geq 0$. Then*

$$\operatorname{Var}_\mu(g) \leq \frac{p-1}{\lambda_p}\int \Gamma(g,g)\,d\mu \tag{2}$$

for every g in the diffusion core (and hence in the closed form domain). In particular, centered L^2 convergence holds with prefactor one and rate at least $\lambda_p/(p-1)$.

Proof. It is enough to work with real-valued core functions. First let h be a mean-zero core function. Raising (1) to the p-th power and taking the right derivative at $t=0$ gives

$$p\int |h|^{p-2}hLh\,d\mu \leq -p\lambda_p\int |h|^p\,d\mu.$$

Invariance and the diffusion chain rule applied to $|h|^p$ give

$$-\int |h|^{p-2}hLh\,d\mu = (p-1)\int |h|^{p-2}\Gamma(h,h)\,d\mu.$$

Thus

$$\lambda_p\int |h|^p\,d\mu \leq (p-1)\int |h|^{p-2}\Gamma(h,h)\,d\mu. \tag{3}$$

Fix a bounded real core function g; the general form-domain statement follows by the standard truncation and closure argument. Put $\alpha = 2/p \in (0,1)$ and, for $\varepsilon > 0$, define the smooth odd function

$$\psi_\varepsilon(s) = s(s^2+\varepsilon)^{(\alpha-1)/2}.$$

It is strictly increasing because

$$\psi_\varepsilon'(s) = (s^2+\varepsilon)^{(\alpha-3)/2}(\varepsilon + \alpha s^2) > 0. \tag{4}$$

The continuous decreasing function

$$c \longmapsto \int \psi_\varepsilon(g-c)\,d\mu$$

has opposite signs at the essential lower and upper bounds of g. Choose a zero c_ε; these centers stay in that bounded interval. Then $h_\varepsilon = \psi_\varepsilon(g-c_\varepsilon)$ has mean zero and belongs to the diffusion core by smooth functional calculus.

Two elementary scalar estimates contain the whole argument. Since $\alpha p = 2$ and $\alpha - 1 < 0$,

$$|\psi_\varepsilon(s)|^p = |s|^p(s^2+\varepsilon)^{(\alpha-1)p/2} \le |s|^2. \tag{5}$$

Also, (4) implies $0 < \psi_\varepsilon'(s) \le (s^2+\varepsilon)^{(\alpha-1)/2}$, whence

$$\begin{aligned} |\psi_\varepsilon(s)|^{p-2}|\psi_\varepsilon'(s)|^2 &\le |s|^{p-2}(s^2+\varepsilon)^{(\alpha-1)p/2} \\ &= \left(\frac{s^2}{s^2+\varepsilon}\right)^{(p-2)/2} \le 1. \end{aligned} \tag{6}$$

Apply (3) to h_ε. The diffusion chain rule and (6) yield

$$\lambda_p \int |\psi_\varepsilon(g-c_\varepsilon)|^p\,d\mu \le (p-1)\int \Gamma(g,g)\,d\mu. \tag{7}$$

Choose a sequence $\varepsilon \downarrow 0$ along which $c_\varepsilon \to c$. Pointwise,

$$|\psi_\varepsilon(g-c_\varepsilon)|^p \longrightarrow |g-c|^2.$$

The bound (5), together with boundedness of g and the centers, permits dominated convergence in (7). Therefore

$$\lambda_p \int |g-c|^2\,d\mu \le (p-1)\int \Gamma(g,g)\,d\mu.$$

Finally, $\mathrm{Var}_\mu(g) = \inf_{a\in\mathbb{R}} \int |g-a|^2 d\mu \le \int |g-c|^2 d\mu$, proving (2).

For completeness, invariance also gives

$$\frac{d}{dt}\,\mathrm{Var}_\mu(P_t g) = -2\int \Gamma(P_t g, P_t g)\,d\mu.$$

Combining this identity with (2) and Gronwall's lemma gives

$$\mathrm{Var}_\mu(P_t g) \le e^{-2\lambda_p t/(p-1)}\,\mathrm{Var}_\mu(g),$$

which is the asserted $K_2 = 1$ conclusion. □

Verification, scope, and novelty status

The included scalar script checks (4), (5), and (6) on 1,159,968 logarithmically distributed triples (p,ε,s); all cases pass. This is a regression check, not a substitute for the displayed proof.

The hypothesis $K_p = 1$ is essential to this route: if a prefactor greater than one is present, differentiation at zero does not yield (3). The proof is specific to diffusions through the exact chain rule and does not assert a pure-jump analogue. It requires no symmetry, duality, or interpolation.

Eight focused attempts are recorded in the accompanying attempt note. The first interpolation and finite-state operator-norm routes retained an uncontrolled K_2. Direct differentiation isolated

(3); the naive fractional power reproduced the source's centering obstruction. The successful upgrade was the shifted smooth fractional power above. Further checks established the energy bound, closure step, quantitative constant, and nonsymmetric L^2 consequence.

Cheap run indexes and bounded web/arXiv searches on 17 August 2026 used the exact source sentence and the phrases $K_p = 1$ *Poincaré nonsymmetric semigroup*, *centered* L^p *exponential convergence Poincaré*, and the weighted formula (3). They found the arXiv and published versions of [1] and later related centered Sobolev results, but no resolution of this exact converse. Novelty confidence is moderate-high pending expert review.

Human-review recommendation. Check the right-differentiation/core step leading to (3) and the routine truncation/closure extension. The scalar regularization and limiting argument are otherwise self-contained.

References

[1] P. Cattiaux, A. Guillin, and C. Roberto, *Poincaré inequality and the* L^p *convergence of semigroups*, Electron. Commun. Probab. 15 (2010), 270–280; arXiv:1003.0784.